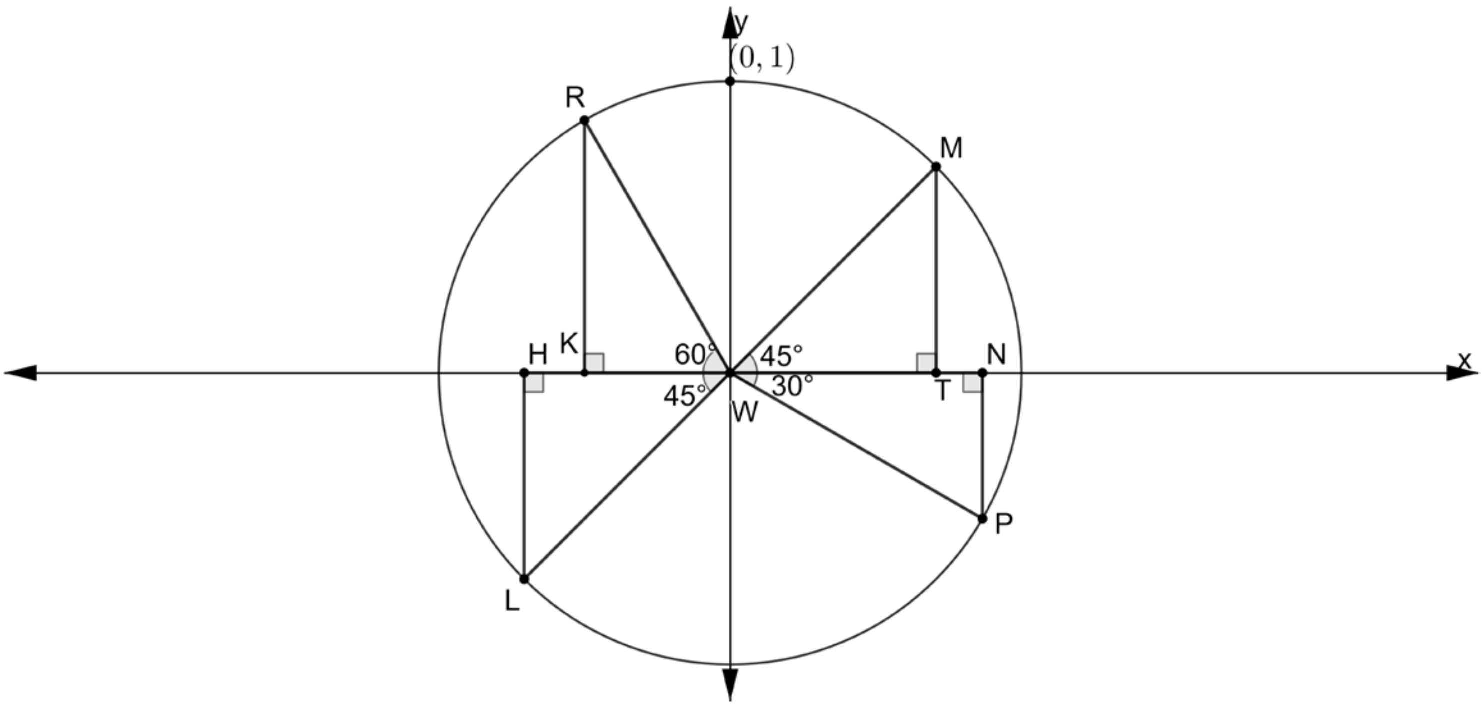
Geometry Name_Answer Key_____
Unit 7 Date_________________
Lesson 7 Practice Period________________

Circle $W$ and right triangles $WMT$, $WPN$, $RWK$, and $LWH$ are shown on the coordinate plane.




1. What is the length of the radius of circle $W$? 1
$(0,0)$ and $(0,1)$ are points.

Complete the table.

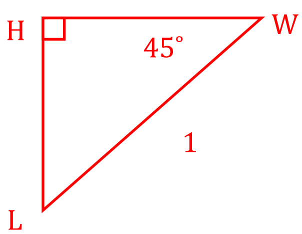
| | Segment | Length |
|---|---|---|
| 2. | $\overline{WM}$ | 1 |
| 3. | $\overline{PW}$ | 1 |

All radii are the same length.

Complete the table.

| | Point | Coordinates |
|---|---|---|
| 4. | $L$ | $(-0.707, -0.707)$ |
| 5. | $P$ | $(0.866, -0.5)$ |
| 6. | $K$ | $(-0.5, 0)$ |
| 7. | $R$ | $(-0.5, 0.866)$ |
| 8. | $N$ | $(0.866, 0)$ |
| 9. | $M$ | $(0.707, 0.707)$ |
| 10. | $H$ | $(-0.707, 0)$ |

Point $L$:

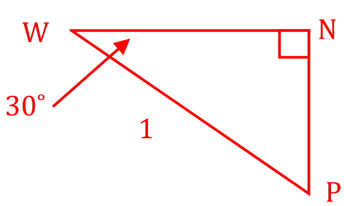



$sin\ 45 = \frac{HL}{1} \rightarrow sin\ 45 = HL \approx 0.707$

$cos\ 45 = \frac{HW}{1} \rightarrow cos\ 45 = HW \approx 0.707$

(point $M$ has same side length of triangles)

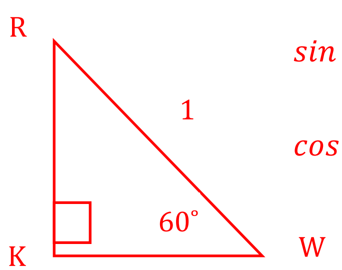
Point $N$ and $P$:




$sin\ 30 = \frac{NP}{1} \rightarrow sin\ 30 = NP \approx 0.5$

$cos\ 30 = \frac{NW}{1} \rightarrow cos\ 30 = NW \approx 0.8660$

Point $K$ and $R$:




$sin\ 60 = \frac{RK}{1} \rightarrow sin\ 60 = RK \approx 0.866$

$cos\ 60 = \frac{KW}{1} \rightarrow cos\ 60 = KW = 0.5$